Deepak Mallampati

APPLIED AI / ML ENGINEER

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | MangaLens (Chrome Web Store) | US-based · F-1 OPT · Open to SF relocation

PROFESSIONAL SUMMARY

Applied AI / ML Engineer with 4+ years building production LLM and ML systems with strong evaluation and data-pipeline discipline. Designs LLM evaluation frameworks (RAGAS, BERTScore, custom metrics), fine-tunes models (QLoRA on H100), and benchmarks with statistical rigor - improving task success by 27% and reducing failed production runs by 60%. Thrives on ill-defined problems, end-to-end ownership, and high-velocity prototyping.

CORE COMPETENCIES & SKILLS

LLM Evaluation & Benchmarking • Fine-Tuning (QLoRA/PEFT) • Scalable Data Pipelines • Statistical Analysis & Experimental Design • Backend Engineering (Python) • MLOps • End-to-End Ownership

LLMs & Evaluation: GPT-4o/5, Claude, Gemini, LLaMA, Qwen | RAGAS, BERTScore, custom domain metrics | A/B Testing | Drift Detection

Fine-Tuning & Training: LoRA/QLoRA/PEFT | PyTorch, TensorFlow | Model training & inference | Instruction-dataset curation

Data & Backend: Python, Java, SQL | FastAPI, REST microservices | PostgreSQL, MongoDB, Redis | Spark, Databricks, Kafka | Pandas, NumPy

MLOps & Cloud: Docker, Kubernetes, Terraform | CI/CD | MLflow, W&B | AWS (SageMaker, Bedrock), Azure, GCP

Methods: Experimental design | Privacy-utility benchmarking | Semantic caching, model routing

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to materially reduce hallucination.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) enabling compliant analytics with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry, reducing failed production runs by **60%**.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets.

Vanguard *Jan 2024 - Jan 2025*

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** on internal eval datasets.
- Engineered safeguards and content controls, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Delivered scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University
Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical readmission dataset (k-anonymity, l-diversity, Laplace differential privacy across four ε levels).
- Reduced re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (RF AUC 0.689 vs 0.696).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated 361-example instruction dataset for controllable long-form generation.

Emerson
Jun 2022 - Apr 2023

ML Trainee

India

- Developed supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Engineered features and end-to-end preprocessing pipelines on raw sensor and operational datasets.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens [Chrome Web Store]

Shipped Chrome extension delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

career-ops

Autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off.

EDUCATION

M.S. in Computer Science - *Kent State University, OH* *2025*

GPA: 3.70 / 4.0